Serat M. Saad

Ph.D. Student in Astronomy, The Ohio State University

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Research Interests

Dynamics: binary stars, radial migration, tests of Modified Newtonian Dynamics (MOND), an alternative to dark matter

Stellar Spectroscopy: chemical abundances, metal-rich stars, star spots, young stellar objects

AI for Astronomy: LLM agents, agent coordination, automation of astrophysical research

Education

Ph.D. in Astrophysics (in progress), The Ohio State University — 2025-Present
Advisors: Prof. Yuan-Sen Ting and Prof. Krzysztof (Kris) Stanek

A.B. in Physics & Applied Mathematics, Vanderbilt University — 2021-2025
Thesis: Understanding Accretion Activity in Young Stars using SDSS-V BOSS Spectra
Advisor: Prof. Keivan Stassun

Honors & Awards

University Distinguished Fellowship, The Ohio State University — 2029-2030
graduate-school fellowship supporting doctoral study

University Fellowship, The Ohio State University — 2025-2027
graduate-school fellowship supporting doctoral study

Hult Prize, National Finalist — 2026
for a student venture expanding access to personalized education

Larry R. Cathey Award, Vanderbilt University — 2025
annual departmental award for the top graduating student studying astronomy

Highest Honors in Astronomy, Vanderbilt University — 2025
for distinction in the undergraduate honors thesis and its defense

International Full-ride Student Grant, Vanderbilt University — 2021-2025
full four-year scholarship covering all costs of undergraduate study

Dean's List, Vanderbilt University — 2021-2025
eight semesters of academic distinction

Buchanan Fellowship, Vanderbilt University — 2022
supported a data-science project applying NLP to British periodical collections

Silver Medal, International Olympiad on Astronomy and Astrophysics — 2020
represented Bangladesh at IOAA 2019 and the Global e-Competition (GeCAA) 2020

Publication Summary

13 papers, 7 as first author, 8 published or accepted in refereed journals.
157 citations from 150 citing papers (SciX Library).

Telescope Time Allocations

Principal Investigator

Large Binocular Telescope, PEPSI, 44 hours · PANTERA: ultra metal-rich stars — 2026B, 2027A
Large Binocular Telescope, PEPSI, 48 hours · PASTA: planet-hosting stars — 2025B, 2026A

Co-Investigator

Large Binocular Telescope, iLocater+PEPSI, 15 hours · close-in binaries (PI: M. Tala Pinto) — 2027A
WIYN, NEID, NN-EXPLORE Key Program · 65 HWO target stars (PI: M. C. Johnson, submitted) — 2026
Automated Planet Finder, Levy, 40 hours · metal-rich stars (PI: D. M. Rowan) — 2026B
Chandra X-ray Observatory, Cycle 26 · Pink Mergers (PI: Prof. Adi Foord) — 2024

Invited Talks and Conference Presentations

Monday Science Seminar, University of Wisconsin–Madison 2026
Title to be announced

LLM Agents Seminar, NASA AI/ML STIG 2026
Hands-on session II: agentic coding and research tools · recording · slides

AAS 245 Winter Meeting, American Astronomical Society 2025
An Alternative to the Supermassive Black Hole Binary Hypothesis for J1430 (Tick Tock)

Cool Stars 22, Cambridge 2024
Using Low Resolution Optical Spectra to Identify Young Stars and Their Properties

IEEE Quantum Computing and Engineering (QCE) 2023
Developing Workforce in Quantum Industry: The Wond'ry Quantum Studio

Teaching & Mentoring

Teaching and grading in astronomy, statistics, data science, and programming, alongside mentoring olympiad students in Bangladesh and graduate-school applicants from underrepresented backgrounds.

Mentor, Project SHORT 2025
free admissions mentoring for underrepresented applicants to graduate and medical programs

Mentor, Bangladesh Olympiad on Astronomy and Astrophysics 2021-2024
national competition that selects and trains Bangladesh's team for the international olympiad (IOAA)

TA, DS 2100: Statistics for Data Science, Vanderbilt University Spring 2025
TA, ASTR 1010: Introductory Astronomy, Vanderbilt University Spring 2024
Grader, DS 1100: Introduction to Data Science, Vanderbilt University Fall 2023
Instructor, Quantum Computing for Everyone, The Wond'ry Vanderbilt Fall 2023
Grader, CS 1100: Introduction to Problem Solving with Python, Vanderbilt University 2022-2023
Tutor, Vanderbilt Student Athletics 2022

Other Experiences

Building an educational AI initiative, communicating new research through writing, and several years training Bangladesh's astronomy olympiad team.

Co-founder, PorAI 2026-Present
educational AI initiative bringing personalized, tutor-like instruction to students and schools

Author and Editor, Astrobites 2026-Present
Journal Referee (A&A) 2026-
Observer, Large Binocular Telescope, 5 nights 2026
Academic Team Member, Bangladesh Olympiad on Astronomy and Astrophysics 2021-2024
Co-lead, The Wond'ry Quantum Studio 2022-2023
student-led quantum computing program at Vanderbilt's innovation center

Team Leader, International Olympiad on Astronomy and Astrophysics 2022
led Bangladesh's national team and served as a question moderator at the international round

First Author Publications and Preprints

7. **Serat M. Saad** & Yuan-Sen Ting (2026)
"More than Half of Recent Astronomy Papers are Written with Language-Model Assistance"
arXiv:2609.10664
6. **Serat M. Saad** & Yuan-Sen Ting (2026)
"Spectroscopic Binary Detection as Agent-Callable Tools: Detecting 40,000+ Main-Sequence Binary Candidates from SDSS DR19 APOGEE Spectra"
The Open Journal of Astrophysics, Vol. 9
5. **Serat M. Saad**, D. M. Rowan, K. Z. Stanek, et al. (2026)
"PANTERA. I. Hunting for the Most Metal-Rich Stars in the Solar Neighborhood with High-Resolution Spectroscopy"
The Open Journal of Astrophysics, submitted
4. **Serat M. Saad** & Yuan-Sen Ting (2026)
"No Gravitational Anomaly in Wide Binaries from Forward Modeling of 3D Orbits"
The Open Journal of Astrophysics, Vol. 9

3. **Serat M. Saad** & Yuan-Sen Ting (2026)
"High-Precision Differential Radial Velocities of C3PO Wide Binaries: A Test of Modified Newtonian Dynamics (MOND)"
The Open Journal of Astrophysics, Vol. 9
2. **Serat M. Saad**, Marina Kounkel, Keivan G. Stassun, et al. (2025)
"ABYSS III: Observing Accretion Activity in Young Stars through Empirical Veiling Measurements"
The Astronomical Journal, Vol. 170, 87
1. **Serat M. Saad**, Kaitlyn Lane, Marina Kounkel, Keivan G. Stassun, et al. (2024)
"ABYSS II: Identification of Young Stars in Optical SDSS Spectra and Their Properties"
The Astronomical Journal, Vol. 167, 125

Co-Author Publications and Preprints

6. Yuan-Sen Ting & **Serat M. Saad** (2026)
"Armillary: A Label-Free Coordinate System for Stellar Spectra"
The Open Journal of Astrophysics, submitted
5. Qinghui Sun, **Serat M. Saad**, Yuan-Sen Ting, et al. (2026)
"Planets Around Solar Twins/Analogs (PASTA) III: Chemical Clock Relations in Planet-Hosting Solar Analogs"
Astronomy & Astrophysics, accepted
4. SDSS Collaboration, including **Serat M. Saad** (2026)
"The Twentieth Data Release of the Sloan Digital Sky Surveys"
The Astrophysical Journal Supplement Series, submitted
3. Zimo Cheng, Sharon X. Wang, Fan Liu, ... **Serat M. Saad**, et al. (2026)
"A Bayesian Search for Planet Engulfment Signatures in Solar Analogs"
AAS Journals, submitted
2. SDSS Collaboration, including **Serat M. Saad** (2026)
"The Nineteenth Data Release of the Sloan Digital Sky Survey"
The Astrophysical Journal Supplement Series, Vol. 285, 9
1. Yuan-Sen Ting, **Serat M. Saad**, Fan Liu, Yuting Shen (2026)
"Egent: An Autonomous Agent for Equivalent Width Measurement"
The Open Journal of Astrophysics, Vol. 9

Non-Refereed Publications

6. **Serat M. Saad** (2026)
"'Total Eclipse of the Heart' in Astronomy"
Astrobites, August
5. **Serat M. Saad** (2026)
"Where Have All the Star Couples Gone?"
Astrobites, August
4. **Serat M. Saad** (2026)
"Cosmic Cannibalism: When Stars Eat Their Planets"
Astrobites, May
3. **Serat M. Saad** (2026)
"Wormholes Might Be More Real Than We Thought"
Astrobites, April
2. **Serat M. Saad** (2026)
"A Wandering Supermassive Black Hole Eating a Star"
Astrobites, March
1. **Serat M. Saad**
"Pixels and Persuasion"
As Seen on TV, digital exhibition, Vanderbilt University Libraries

References

Krzysztof (Kris) Stanek, The Ohio State University — Ph.D. Advisor, stanek.32@osu.edu
University Distinguished Scholar; Guggenheim Fellow (2022)

Yuan-Sen Ting, The Ohio State University — Ph.D. Advisor, ting.74@osu.edu
Inaugural Chair, NASA AI/ML Science and Technology Interest Group; Kavli Scholar (2026)

Keivan G. Stassun, Vanderbilt University — Undergraduate Advisor, keivan.stassun@vanderbilt.edu
National Medal of Science (2024); MacArthur Fellow (2024)